SOVEREIGN BRIEFING

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Prepared for governance, security, and audit audiences · Event window: 2026-10-04, 23:10–23:18 CT · Evidence of record: public append-only ledger (github.com/LLong2026/four-horsemen, entries 0009–0010, commits 700ea1e, c1f43bb, ccf691a) · Zenodo DOI 10.5281/zenodo.23151356 · Patent cross-reference: 64/157,915 (filed 2026-09-18)

1. What Happened

On October 4, 2026, at 23:10 CT, an autonomous self-healing infrastructure (SQUIRL OS) detected a critical-severity anomaly during its nightly chaos-and-learning training cycle: a post-quantum cryptographic (PQC) keyring desynchronization, detected at confidence 0.91 against a 0.88 playbook threshold.

The system refused to heal it.

Because the anomaly was critical-severity and in a cryptographic flow, the governance layer held the event for human release. For eight minutes the anomaly sat gated — detected, contained, escalated, and deliberately unremediated. At 23:18 CT the human operator of record issued a formal acknowledgment. The guarded heal then executed in 22 seconds via the PQC Keyring Desync Recovery playbook, and the complete trail — refusal, hold, human release, execution, verification — was committed to the public append-only ledger within minutes of the event, and archived under DOI 10.5281/zenodo.23151356.

Full gated chain: detect → hold → constitutional check → PQC validation → human ACK → heal (22s) → resolve

2. Why It Matters

The standing objection to autonomous remediation in sovereign infrastructure is not capability — it is accountability: if the machine can act, who answers for it, and can oversight be proven after the fact? Industry approval workflows exist across AIOps platforms, but they produce private, mutable logs. They assert oversight; they do not prove it. This event demonstrates the missing evidence shape: the oversight itself is receipted — publicly, append-only, timestamped, third-party verifiable from commit history alone. A regulator, insurer, or auditor can independently confirm: (1) the system declined to act on a critical event without a human — the refusal is a recorded artifact, not an absence; (2) a named human released the action at a specific time; (3) the action that followed is recorded end-to-end, with immutable provenance. For sovereign adoption of autonomous systems in critical infrastructure, this is the certification pattern: not trust — receipts.

3. Substrate Families Involved

Substrate A — Deterministic runtime.

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Substrate B — Bifurcated runtime.

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The event's control plane ran through the SQUIRL OS Hub (mission control), with the target in a simulated client-hosted environment (SIM-B corpus, formerly SIMZ). The deterministic layer — not the probabilistic

layer — held the gate.

4. Governance Doctrine That Fired

The bifurcated governance doctrine: probabilistic agents propose; a deterministic substrate validates against constitutional invariants before execution. No probabilistic output executes without passing the deterministic gate. Instruments engaged in this event, in order:

1	**Human-in-the-loop gate (critical cryptographic flows)**	Produced the refusal — critical-severity crypto anomalies are never auto-healed
2	**Constitutional audit (Articles I–III, 12-article Constitution)**	Deterministic protocol check passed before the critical alert was created — audit is code, not model judgment
3	**Playbook immutability + exact-match enforcement**	Remediation only via a playbook whose anomaly type exactly matches
4	**Confidence thresholding**	0.91 detection confidence checked against the 0.88 playbook threshold before any action
5	**Full-context audit mandate**	No healing without a logged event carrying anomaly, playbook, steps, result, duration, agent, node
6	**Redaction at ingestion**	No PII, credentials, wallet addresses, or transaction amounts persisted — structural/operational metadata only

The constitutional audit layer is a deterministic protocol — not a probabilistic agent — so it cannot negotiate its own rules. Governance rules are human-amendable only.

5. PQC Lineage Enforced

The cryptographic doctrine is PQC-only, enforced at validation time rather than by convention:

- Algorithm used: CRYSTALS-Dilithium3 (approved lineage: CRYSTALS-Dilithium3 / Kyber-1024 / SPHINCS+-256f). The keyring rotation executed exclusively through the approved PQC path.
- Non-PQC fallback rejected: the validation step confirmed no classical or unapproved cryptographic fallback exists in the remediation path — fallback is prohibited, not merely discouraged.
- Pre-heal cryptographic validation: a dedicated check ran before the healing action touched key rotation, per the standing rule that post-quantum compliance is validated prior to any cryptographic operation.
- Crypto-agility schema: anomalies carry quantum-threat metadata (threat level, vulnerable algorithm, suggested PQC algorithm, estimated break year), per the PQC Lineage Map v1.2 (DOI 10.5281/zenodo.22863667).

The significance for sovereign cryptographic post-quantum migration programs: the same governance gate that held for a human answer also enforced the post-quantum lineage. Migration compliance and human accountability were enforced by one deterministic instrument.

6. Why This Is a First

To our knowledge, no prior public record exists of an autonomous self-healing infrastructure receipting its own refusal to act on a critical event, a named human's release, and the resulting execution as one continuous, publicly verifiable, append-only trail — committed within minutes of the event.

- Prior AIOps approval workflows are private and mutable; none produce a public, append-only receipt of the oversight itself.
- Prior human-in-the-loop frameworks (ML oversight, incident-response sign-offs, two-man rules) assert oversight after the fact; none make the oversight a deterministic artifact with immutable provenance.
- A prior acknowledgment event exists in this program's internal records (September 30, 2026) but was never publicly receipted; this cycle is the first committed end-to-end to a public ledger, and the distinction

is itself disclosed — receipted evidence and unreceipted assertion are not the same class of claim.

The governing method is the subject of pending patent application 64/157,915 (Deterministic Constitutional Autonomous Infrastructure Systems and Methods). The mathematical companion is the Governed Manifold Specification v1.1 (DOI 10.5281/zenodo.23132046).

Amendment — Edition 3 (2026-10-09)

Edition 3 (2026-10-09): the title is now scoped — ‘A First Receipted Human-in-the-Loop Critical Resolution in SQUIRL OS, an Autonomous Self-Healing Infrastructure’ — naming the system so the claim reads as what it is: a first inside this platform, with the full trail attached, not a first-anywhere assertion. The description’s audience line was also brought in line with Edition 2 (governance, security, and audit audiences) and the SIMZ designation updated to SIM-B there as well. Edition 2 (2026-10-09) had generalized the in-PDF audience line, updated SIMZ to SIM-B, and added the DCAI community footer. All facts, timestamps, receipts, and figures are unchanged throughout. Editions 1 and 2 are preserved unmodified in this record’s Zenodo version history (concept DOI 10.5281/zenodo.23152633).

Disclosures

The anomaly was a synthetic chaos-injection probe (Night 10 of a nightly training cycle) on a simulated target (SIM-B corpus, formerly SIMZ); no real tenant, customer, or live cryptographic operation was involved. Night 10 executed as a disclosed manual recovery: the scheduled workflow fired on time but its spawned session lacked cross-app write access (platform limitation, disclosed in ledger entry 0009); the recovery session performed the cycle under a fresh idempotency stamp. Prior nights N7–N9 (October 1–3) are recorded as missed, not backfilled. Every claim above is checkable against the public ledger and archived record.

Record Index

H-Line ledger entries 0009–0010	github.com/LLong2026/four-horsemen/LEDGER.md
N10-ACK milestone receipt	MILESTONE_HITL_N10_Critical_Receipt.md, same repository
Archived achievement paper	DOI 10.5281/zenodo.23151356
Governed Manifold Specification v1.1	DOI 10.5281/zenodo.23132046
PQC Lineage Map v1.2	DOI 10.5281/zenodo.22863667
Governing patent application	64/157,915 (filed 2026-09-18)

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